

Bacterial Meningitis in Infants



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KEYWORDS

- Neonatal bacterial meningitis • Very low birth weight • Lumbar puncture
- Cerebrospinal fluid • Antibiotics • Vaccine

KEY POINTS

- Neonatal bacterial meningitis is uncommon but is associated with high mortality and morbidity.
- Group B *Streptococcus* (GBS) is the most common cause of neonatal meningitis.
- *Escherichia coli* has recently become the most common pathogen isolated from very low birth weight infants with meningitis.
- Infants with culture-proven meningitis can have negative blood cultures and normal cerebrospinal fluid parameters.
- All infants showing signs of infection and with suspected early-onset or late-onset sepsis should undergo a lumbar puncture.
- A GBS vaccine for the prevention of neonatal GBS disease including meningitis is currently under development.

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INTRODUCTION

Bacterial meningitis is a serious infection associated with high mortality and morbidity in the neonatal population. Prompt diagnosis and treatment are essential to achieving good outcomes in affected infants. Although overall incidence and mortality have declined over the last several decades, morbidity associated with neonatal meningitis remains virtually unchanged.^{1,2} Prevention strategies, adjunctive therapies, and improved diagnostic strategies have been the focus of recent research seeking to improve the outcomes.³

DESCRIPTION OF THE DISEASE

Meningitis is the acute inflammation of the meninges, subarachnoid space, and brain vasculature resulting from infection.⁴ Neonatal meningitis is categorized as early and late onset, which is defined by the presence of signs of infection and organism isolation from cerebrospinal fluid (CSF) cultures at less than or equal to 72 hours and greater than 72 hours of life, respectively.^{3,5-7}

EPIDEMIOLOGY

The incidence of neonatal meningitis varies by geographic location (Fig. 1).⁸⁻¹⁰ Compared with older age groups, the incidence of meningitis is highest during the neonatal period.^{9,16}

Developed Countries

The incidence of culture-proven neonatal meningitis is estimated at 0.3 per 1000 live births in developed countries.^{2,10,16} However, this number is likely an underestimation of the true incidence. For infants in the intensive care nursery who are evaluated for sepsis, 30% to 50% do not have a lumbar puncture (LP) performed.^{6,17} When an LP is performed, more than 75% of the time it occurs after the initiation of antibiotics, possibly biasing CSF culture results.^{6,17,18}

In developed countries, mortality from neonatal meningitis ranges from 10% to 15%.^{2,8} In a prospective study including 444 cases of confirmed meningitis from 2001 to 2007, mortality in premature infants compared with term infants was more

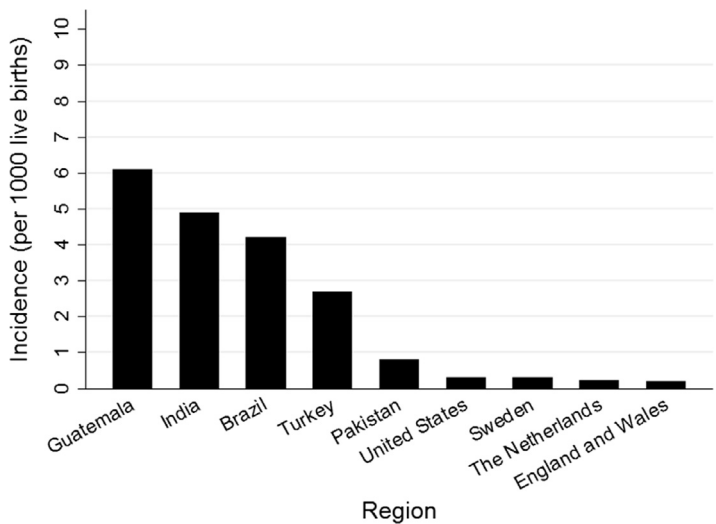


Fig. 1. Incidence of neonatal meningitis across the world. (Data from Refs.^{2,11-15})

than 2-fold higher (26% vs 10%, $P<.01$).⁸ Up to 50% of infants with a history of meningitis are neurologically impaired, with 25% having severe disability.^{2,8,19} With advances in medical practices, the incidence and mortality associated with meningitis have declined over the past 40 years; however, morbidity remains unchanged.¹⁹

Developing Countries

In developing countries, the reported incidence of neonatal meningitis is much higher at 0.8 to 6.1 per 1000 live births, with a mortality of 40% to 58%.^{9,11} True values may be higher because of underreporting in regions with limited resources, diagnostic testing, and access to health care.⁹

CAUSE

The types and distribution of organisms commonly observed in neonatal meningitis depend on postnatal age, location, and gestational age. The distribution of organisms seen in neonatal meningitis is similar to that of neonatal sepsis (Table 1).^{1,6,16}

Early-onset Meningitis

Despite the institution of maternal intrapartum prophylaxis, group B *Streptococcus* (GBS) has remained the most common cause of neonatal sepsis and meningitis since the early 1980s, responsible for greater than 40% of all early-onset infections.^{2,6,20} *Escherichia coli* is the second most common pathogen and is isolated in 30% of all early-onset infections.⁶ Since the 1990s, *E coli* has emerged as the most common cause of early-onset sepsis and meningitis among very low birth weight (VLBW; ie, <1500 g birth weight) infants.^{21–24}

Late-onset Meningitis

Late-onset meningitis is predominantly seen in premature infants, and the incidence is directly related to decreasing birth gestational age and weight.²⁵ Surveillance of 6956 VLBW infants from 1998 to 2000 found coagulase-negative staphylococci (48%) and *Staphylococcus aureus* (8%) to be the first and second most common pathogens, respectively.⁷ *E coli* (5%) and *Klebsiella* (4%) spp were the most common gram-negative causes of late-onset infections.^{5,7} Although GBS (2%) was less common in this cohort, other studies found that infants were more likely to have confirmed meningitis with late-onset GBS sepsis compared with early-onset GBS sepsis (Table 2).^{7,26,27}

Table 1

Common pathogens of meningitis and commonly used empiric antibiotics

Type	Major Pathogens	Empiric Antibiotics
Early onset	Group B <i>Streptococcus</i>	Ampicillin
	<i>Escherichia coli</i>	Gentamicin
	<i>Listeria monocytogenes</i>	Cefotaxime
	<i>Streptococcus pneumoniae</i>	
Late onset	Coagulase-negative <i>Staphylococcus</i>	Vancomycin
	<i>Staphylococcus aureus</i>	Gentamicin
	<i>E coli</i>	Cefotaxime
	<i>Klebsiella</i> spp	Ampicillin
	<i>Enterococcus</i> spp	
	<i>Enterobacter</i> spp	
	<i>Pseudomonas</i> spp	
	Group B <i>Streptococcus</i>	

Table 2 Infants with late-onset versus early-onset GBS sepsis complicated by meningitis					
N (Ref.)	Years	Study Population	Proportion of Infants with Late-onset GBS Sepsis Complicated by Meningitis, n (%)	Proportion of Infants with Early-onset GBS Sepsis Complicated by Meningitis, n (%)	P
347 ²⁶	2001–2003	GA 23–43 wk PNA ≤90 d Confirmed GBS sepsis	84 of 136 (62)	33 of 206 (16)	<.001
179 ²⁷	1997–2004	GA 31–40 wk Confirmed GBS sepsis	13 of 24 (54)	22 of 155 (14)	<.01

Abbreviations: GA, gestational age; PNA, postnatal age.

PATHOGENESIS

Although several mechanisms in the development of neonatal meningitis have been described, primary bloodstream infection with secondary hematogenous distribution to the central nervous system (CNS) is the most common (Box 1).¹⁶ For this reason, the epidemiology and microbiology of neonatal meningitis are similar to those of neonatal sepsis.¹

Early-onset Infection

Organisms present in the maternal genitourinary tract ascend from the vagina and can infect the amniotic fluid through disruptions in the amniotic membranes, which the infant then aspirates.²⁸ Organisms can also colonize exposed neonatal skin and mucosa during passage through the birth canal and invade through barrier disruptions.¹ Organisms such as *Listeria monocytogenes* can also be transmitted transplacentally.¹⁶ In rare cases, hematogenous transmission of GBS from maternal bacteremia has been reported as a cause of early-onset GBS infections in infants.²⁹

Late-onset Infection

Organisms can be acquired from the colonized mother, as seen with GBS.^{30,31} Poor hand hygiene among caregivers and hospital staff can result in the transfer of organisms between infected and uninfected infants.³² Foreign, invasive devices such as ventricular reservoirs, ventricular shunts, endotracheal tubes, venous or arterial catheters, urinary catheters, and feeding tubes can also introduce pathogens to the infant.¹⁶ Exposure to prolonged courses of empiric antibiotics for suspected infections can also result in increased risk for late-onset infections.³³ Among 365 VLBW infants

Box 1

Mechanisms for development of neonatal meningitis

1. Primary bloodstream infection with secondary hematogenous spread to CNS

2. Presence of an infectious focus with secondary bloodstream infection and hematogenous spread (eg, osteomyelitis)

3. Presence of an infectious focus with direct extension into the CNS (eg, sinus infection)

4. Primary CNS infection resulting from disruptions caused by head trauma, neurosurgery, or congenital defects (eg, myelomeningocele)

less than or equal to 32 weeks' gestational age, infants exposed to empiric antibiotics for greater than or equal to 5 days had increased odds of developing late-onset sepsis (odds ratio, 2.45; 95% confidence interval, 1.28–4.67).³⁴

Infection of the Central Nervous System

After attaching to the endothelium of the cerebral microvasculature and choroid plexus, bacteria can enter the CSF by several mechanisms (**Box 2**).^{1,4,35} Inflammatory mediators are then released into the CSF in response to the presence of bacterial products, resulting in meningitis and increased permeability of the blood-brain barrier.^{1,31}

RISK FACTORS

Risk factors for the development of neonatal meningitis are similar to those for neonatal sepsis (**Box 3**).^{5,10,36} Immaturity of the neonatal immune system, impaired phagocytic ability of neutrophils and monocytes, and diminishing maternal antibodies all contribute to increased risk of infections in both term and preterm infants.^{4,24} Because most maternal immunoglobulins do not cross the placenta before 32 weeks' gestation, infants born extremely preterm are at significantly higher risk for infections.¹⁰ However, early initiation of breastfeeding may be protective against infections because of transfer of immunoglobulin A.³⁷

CLINICAL PRESENTATION

The clinical signs of neonatal meningitis can be subtle and nonspecific (**Box 4**).^{16,19,38} Meningitic signs such as convulsions, irritability, bulging fontanel, and nuchal rigidity are often late findings that are associated with poor outcomes.^{2,10,38}

Exposure to intrapartum antibiotic prophylaxis (IAP) against GBS has led to concerns that the signs of neonatal infections could be delayed or masked. Several studies have determined no significant difference in the clinical presentation of early-onset GBS disease between infected infants with and without prior exposure to IAP.²⁰ Signs of early-onset sepsis manifested in 90% of infected infants within the first 24 hours of life.²⁰

DIAGNOSIS

To confirm the diagnosis of neonatal meningitis, an LP is needed to collect CSF. Positive growth on the CSF culture provides identification of the organism and enables refinement of therapy.^{2,39} The LP is frequently deferred during the septic work-up because of concerns of exacerbating clinical deterioration in the sick infant.^{39–41}

Performing or Deferring the Lumbar Puncture

Because the LP is an invasive procedure with risks, it is difficult to determine which infant should receive one as part of the septic work-up.^{40,41} Among infants with positive blood cultures, up to 30% have a concurrent positive CSF culture.⁴² However, in

Box 2

Bacterial mechanisms for entry into the CSF

1. Transcellular movement across the endothelial cell (eg, GBS, *E coli*)
2. Paracellular movement by disruption of intercellular tight junctions
3. Transport across the blood-brain barrier and blood-CSF barriers within phagocytes (eg, *L monocytogenes*)

Box 3**Risk factors for neonatal meningitis**

1. Prematurity
2. Maternal rectovaginal GBS colonization
3. Chorioamnionitis or maternal fever
4. Premature rupture of membranes
5. Prolonged rupture of membranes (>18 hours)
6. Invasive fetal monitoring
7. VLBW (<1500 g)
8. Prolonged hospitalization
9. Presence of external devices (eg, reservoirs, shunts, catheters)

infants with confirmed meningitis, 15% to 38% have a negative blood culture.^{27,43–45} In rare cases, the blood and CSF cultures can be discordant.⁴⁴ Approaches in which only infants with confirmed bacteremia are evaluated for meningitis result in missed diagnoses of meningitis.

The incidence of meningitis among asymptomatic infants with risk factors is very low (<1%).^{46,47} When clinical signs can be attributed to noninfectious causes, such as respiratory distress syndrome or transient tachypnea of the newborn, clinical judgment is required in deciding when to perform an LP.⁴³ Among 238 infants admitted for respiratory distress without other symptoms, 17 of 238 (7%) infants had a positive blood culture, and none had meningitis.⁴⁸

The current recommendation is to perform an LP on all clinically stable infants suspected to have early-onset or late-onset sepsis and who are showing signs of

Box 4**Clinical signs of neonatal meningitis**

1. Fever or hypothermia
2. Irritability or lethargy
3. Hypotonia
4. Feeding intolerance or vomiting
5. Respiratory distress
6. Apnea
7. Bradycardia
8. Hypotension
9. Poor perfusion
10. Seizures
11. Bulging anterior fontanel
12. Nuchal rigidity
13. Jaundice
14. Hypoglycemia or hyperglycemia
15. Diarrhea

infection.^{20,39,43} Whenever possible, the LP should be performed before the administration of antibiotics.

Interpreting Cerebrospinal Fluid Parameters

Infants are often exposed to intrapartum or empiric antibiotics before receiving an LP, which can result in falsely negative CSF cultures in those with meningitis.¹⁸ In these cases, CSF parameters are used to help determine the likelihood of meningitis.

CSF indexes vary according to age, with normal values in infants poorly defined.^{16,44,45} Common practice in infants states that a CSF leukocyte count greater than or equal to 20/mm³ suggests bacterial meningitis.¹⁶ A study involving 1064 infants showed that infants less than 28 days old without bacterial meningitis had a median CSF leukocyte count of 3/mm³ and 95th percentile value of 19/mm³.⁴⁹ However, in a study evaluating 9111 infants at greater than or equal to 34 weeks' gestational age, the use of 20/mm³ as a cutoff resulted in a missed diagnosis in 13% of infants with confirmed meningitis.⁴⁴ Several infants with confirmed meningitis had normal CSF parameters without bacteremia. CSF protein and glucose were considered poor predictors of meningitis because of considerable overlap of values between infants with and without confirmed meningitis. A study of 4632 infants of less than 34 weeks' gestational age reached similar conclusions in the preterm population.⁴⁵ There is great difficulty in predicting the diagnosis of meningitis solely based on CSF parameters, so CSF culture continues to be the gold standard for diagnosis.

Ancillary Tests

The polymerase chain reaction (PCR) has been explored as a diagnostic tool for meningitis. In addition to improved sensitivity and specificity, PCR also allows quicker detection of pathogens compared with traditional cultures.⁵⁰ A real-time PCR assay designed to detect multiple pathogens including *Streptococcus pneumoniae*, *E coli*, GBS, *S aureus*, and *L monocytogenes* had an overall higher detection rate of any CSF pathogen compared with traditional cultures (72% vs 48%).⁵¹ Among patients exposed to antibiotics before collection of CSF, PCR had a higher detection rate compared with culturing (58% vs 29%).⁵¹ Further testing is needed before PCR can be used routinely in the diagnosis of bacterial meningitis.⁵²

Other tests used to aid in clinical decision making include the complete blood count with differential and C-reactive protein. Studies examining the usefulness of these tests in the diagnosis of neonatal bacterial meningitis are limited (**Table 3**).

Repeating the Lumbar Puncture

The need for a repeat LP during treatment in an infant with confirmed meningitis has been debated. Some experts recommend routinely repeating an LP in all patients at 48 hours, whereas others suggest repeating an LP only if clinical conditions are not improved by 24 to 72 hours after beginning therapy.^{56–59}

In a retrospective study of 14,018 infants, 221 infants were identified as having culture-positive meningitis, with 118 infants (53%) receiving 2 or more LPs during the treatment course.⁵⁶ Among infants with available mortality data receiving 2 or more LPs, 6 of 23 (26%) infants with repeat positive cultures died compared, with 6 of 81 (7%) infants with repeat negative cultures ($P = .02$). No significant difference in mortality was seen among the 81 (7%) infants with a repeat negative culture compared with the 90 (12%) infants with meningitis with no repeat LP ($P = .32$). A survey of 109 pediatricians and neonatologists across the northwest England found that 89 (82%) practitioners did not routinely repeat the LP in infants with bacterial meningitis unless clinically indicated.⁵⁷

Table 3
Available studies on biomarkers for neonatal meningitis

Biomarker	N (Ref.)	Study Population	Notable Findings
Complete blood count differential ratio $\left(\frac{(\% \text{ lymphocytes} + \% \text{ monocytes})}{(\% \text{ polymorphonuclear leukocytes} + \% \text{ band forms})} \right)$	72 ⁵³	Term PNA <4 wk	Complete blood count differential ratio <1.5 as cutoff for predicting bacterial meningitis, test achieved the following: Sensitivity = 100% Specificity = 67% Positive predictive value = 47% Negative predictive value = 100%
	40 ⁵⁴	Term PNA <28 d	Used same cutoff values as previous study on different cohort achieved the following: Sensitivity = 70% Specificity = 54% Positive predictive value = 39% Negative predictive value = 81%
Peripheral white blood cell count	8312 ⁴⁴	GA ≥34 wk PNA 0–150 d	Use as predictor for bacterial meningitis had positive likelihood ratio <1.0 Neither sensitive nor specific
C-reactive protein, CSF	23 ⁵⁵	GA 28–41 wk PNA 0–6 wk	1 of 7 infants without infection, 0 of 5 infants with sepsis without meningitis, and 2 of 11 infants with meningitis had C-reactive protein >1 mg/dL Levels do not distinguish between infants with meningitis and those with no confirmed infection

TREATMENT

Antimicrobial Therapy

Prompt initiation of antibiotics is critical. Delays in treatment are associated with increased mortality and morbidity.⁶⁰ Empiric antimicrobials used in suspected meningitis require adequate CSF penetration and sensitivity against the most probable pathogens.^{10,60} On identification of the pathogen and its susceptibilities, antimicrobial coverage should be adjusted accordingly (Table 4).

Duration of Antimicrobial Therapy

For uncomplicated meningitis, the minimum recommended treatment durations are the following^{10,61,64}:

- Fourteen days for GBS, *L monocytogenes*, and *S pneumoniae*
- Twenty-one days for *Pseudomonas* and gram-negative enteric bacteria such as *E coli*

Longer treatment courses are recommended for infants with meningitis with delayed clinical improvement after beginning therapy or with complications such as brain abscesses, ventriculitis, or brain infarctions.⁶²

Antibiotic (Ref.)	Susceptible Bacteria	Notes
Penicillin G ⁶¹	GBS	Monotherapy acceptable if GBS confirmed by culture and clinical improvement is observed
Ampicillin ^{2,6,62,63}	GBS <i>L monocytogenes</i> <i>Enterococcus</i> sp	17%–78% of <i>E coli</i> isolates resistant Poor CNS penetration Requires higher doses for meningitis
Gentamicin ^{2,62,64}	<i>E coli</i> <i>Klebsiella</i> sp <i>Enterobacter</i> sp <i>Pseudomonas</i> sp <i>Citrobacter</i> sp <i>Serratia</i> sp	Poor CNS penetration Synergistic effect with ampicillin in treatment of <i>L monocytogenes</i> <i>Pseudomonas</i> sp may require combination therapy with a second agent Requires therapeutic drug monitoring
Cefotaxime ^{24,62,64}	<i>E coli</i> <i>Klebsiella</i> sp <i>Enterobacter</i> sp <i>Citrobacter</i> sp <i>Serratia</i> sp	Good CNS penetration Used instead of gentamicin in cases of suspected or confirmed meningitis Not active against <i>L monocytogenes</i> or <i>Enterococcus</i> sp
Meropenem ^{64,65}	<i>E coli</i> <i>Klebsiella</i> sp <i>Enterobacter</i> sp <i>Citrobacter</i> sp <i>Serratia</i> sp <i>Pseudomonas</i> sp	Good CNS penetration Limit use to multidrug-resistant organisms (eg, extended-spectrum β -lactamase-producing organisms)
Vancomycin ^{62,66}	Coagulase-negative staphylococci <i>S aureus</i> <i>Enterococcus</i> sp	Variable CNS penetration Effective against methicillin-resistant <i>S aureus</i> Requires therapeutic drug monitoring
Nafcillin ^{66,67}	Methicillin-sensitive <i>S aureus</i>	Good CNS penetration Superior to vancomycin for treatment of methicillin-sensitive <i>S aureus</i>

Adjunctive Therapy

In an effort to improve outcomes in infants with meningitis, several adjunctive therapies have been explored, including the use of intraventricular antibiotics, dexamethasone, intravenous immunoglobulins, granulocyte or granulocyte macrophage colony-stimulating factor, and oral glycerol.² At present, none of the proposed adjunctive therapies are used in routine practice.

LONG-TERM OUTCOMES

Survivors of neonatal meningitis are at considerable risk for long-term neurologic impairment.^{68,69} A prospective study that followed 1717 survivors of neonatal meningitis to 5 years of age found that children who had neonatal meningitis were 10 times more likely to have moderate or severe disability than those who never had meningitis.⁶⁸ Certain characteristics can help identify infected infants at the highest risk for a poor outcome (Box 5).^{70,71}

PREVENTING NEONATAL MENINGITIS
Intrapartum Antibiotic Prophylaxis

Since the adoption of IAP use in 1996 and universal antenatal screening for GBS colonization in 2002, the incidence of early-onset GBS infections has decreased significantly, from 1.8 cases per 1000 live births in 1990 to 0.26 cases per 1000 live births in 2010.⁷² The incidence of late-onset GBS disease remains unaffected by IAP use.^{30,72}

IAP use has also been implicated in the increased proportion of non-GBS early-onset infections seen in VLBW infants. Since the late 1990s, *E coli* has surpassed GBS as the predominant pathogen observed in early-onset infections among VLBW infants.^{6,21–23} This finding likely reflects the success of IAP in reducing the incidence of early-onset GBS infections rather than an increase in the incidence of non-GBS infections.⁷³ This change in proportion of non-GBS early-onset infections has not been observed in term infants.⁶

Reports from single-center studies have associated frequent use of ampicillin for IAP with an increase in ampicillin-resistant *E coli* infections, particularly in premature infants.^{22,74} However, rates of ampicillin-resistant *E coli* have also increased in the general community.²⁰ Furthermore, a multicenter trial involving 389 infants with confirmed early-onset sepsis found that, between infants exposed and not exposed to IAP, frequencies of ampicillin-resistant *E coli* infections were not significantly different.⁶ Although GBS has become increasingly resistant to clindamycin and erythromycin, sensitivity to ampicillin remains unchanged.⁷⁵

Group B Streptococcus Vaccine

Vaccines against GBS can reduce the number of missed opportunities for screening and IAP administration caused by false-negative screens, precipitous deliveries, or

Box 5
Predictors of poor neurologic outcomes in survivors of bacterial meningitis

1. Seizures lasting more than 72 hours
2. Presence of coma
3. Hypotension requiring inotropic support
4. White blood cell count less than or equal to $5000 \times 10^9/L$
5. Abnormal electroencephalogram findings

Table 5
GBS vaccine clinical trials

ClinicalTrials.gov Identifier (Ref.)	Phase	Objectives	Vaccine	Subjects	N	Design	Study Period
NCT00645346 ⁷⁷	I	Safety, tolerability, and immunogenicity	GBS glycoconjugate	Healthy nonpregnant women aged 18–40 y	130	Randomized, single center, single blind, placebo controlled	2008–2009
NCT01193920 ⁷⁸	Ib/II	Safety and immunogenicity	Trivalent GBS	Healthy nonpregnant and pregnant women aged 18–40 y	380	Randomized, single center, single blind, placebo controlled	2010–2012
NCT01446289 ⁷⁹	II	Immune response; amount of vaccine-induced antibody transferred to infant	Trivalent GBS	Healthy pregnant women aged 18–40 y	86	Randomized, multicenter, single blind, placebo controlled	2011–2013
NCT02046148 ⁸⁰	II	Safety and immunogenicity; placental transfer of GBS antibodies; levels of GBS antibodies in infants; levels of GBS antibodies in breast milk	Trivalent GBS	Healthy pregnant women aged 18–40 y	75	Randomized, multicenter, double blind, placebo controlled	2014–2015

extremely preterm births.^{6,72} Maternal immunity to the most common serotypes of GBS can be transferred passively to the infant and protect against early-onset and late-onset infections.⁷⁶ A trivalent GBS vaccine has shown promise in phase I and II trials, with another phase II trial currently recruiting participants (Table 5).⁷⁶

SUMMARY

Neonatal meningitis is a serious disease that requires a high index of suspicion, prompt diagnosis, and rapid treatment. Although the incidence and mortality have declined with improved neonatal intensive care practices and universal adoption of preventative screening and prophylaxis programs, the associated morbidity remains unchanged. Performing an LP to collect CSF is critical to confirming the diagnosis, determining the causative pathogen, and refining antimicrobial therapy. Through better diagnostic practices and development of vaccines, there is hope that the burden of this dangerous disease may be further reduced.

Best practices box

What is the current practice?

Neonatal bacterial meningitis

Best practice/guideline/care path objectives

- Promptly diagnose with performance of LP
- Begin antibiotics without delay
- Reduce mortality and prevent long-term neurodevelopmental impairment

What changes in current practice are likely to improve outcomes?

- Collection of CSF in all stable, symptomatic infants suspected of early-onset or late-onset infections with additional signs beyond respiratory distress
- Increased vigilance in identifying and treating mothers with GBS colonization
- Introduction of an effective GBS vaccine in the prevention of neonatal GBS disease

Major recommendations

- Perform a LP on all clinically stable infants with suspected sepsis and meningitis (grade 1B)
- Begin empiric antibiotics in all cases of suspected sepsis and meningitis (grade 1A)
 - For suspected early-onset meningitis, consider ampicillin and gentamicin or cefotaxime in infants
 - For suspected late-onset meningitis, consider vancomycin in addition to ampicillin and gentamicin or cefotaxime
- Repeat the LP in infants who fail to show clinical improvement 24 to 48 hours after initiation of antibiotics (grade 1B)
- Repeat LPs are unnecessary in infants who show rapid clinical improvement after initiation of antibiotics and at end of successful therapy (grade 1B)
- All infants with a history of bacterial meningitis should be followed long term for development of neurologic sequelae (grade 1A)

Summary statement

A high index of suspicion, prompt diagnosis, and rapid initiation of antibiotics are essential to reducing mortality and morbidity associated with neonatal bacterial meningitis.

Data from Refs. ^{2,10,20}

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